

Multi-Task Learning with Vision Transformers: Joint Classification and Saliency Prediction

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Why Multi-Task Learning?

- Most deep learning models do one thing at a time: classify, detect, or segment.
- Humans solve many visual tasks simultaneously using shared representations.
- **Multi-task learning** trains one model on multiple tasks, encouraging a shared backbone to learn features useful for each.
- Potential benefits: better generalization, more efficient use of data and compute.

Our question: Can a single Vision Transformer classify images *and* predict where humans look?

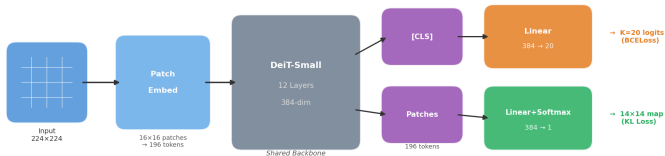
Why Saliency?

- **Saliency prediction** estimates where people look when viewing an image.
- Related to classification: object recognition and visual attention share low-level features (edges, textures, object boundaries).
- The SALICON dataset provides dense fixation maps for COCO images, enabling supervised training.
- Pairing classification labels with saliency ground truth lets us study how the two tasks interact inside a shared model.

Vision Transformer (DeiT-Small)

- **DeiT-Small:** a data-efficient Vision Transformer pretrained on ImageNet.
- Splits each 224×224 image into a grid of 16×16 patches $\rightarrow$ 196 patch tokens + 1 CLS token.
- 12 transformer layers, 384-dimensional embeddings, 22M parameters.
- We fine-tune the entire backbone on our paired dataset.

Multi-Task Architecture



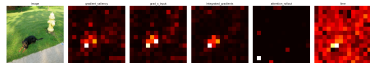
- **Classification head:** Linear layer on CLS token $\rightarrow$ 20 category logits.
- **Saliency head:** Linear layer on each patch token $\rightarrow$ softmax $\rightarrow 14 \times 14$ probability distribution.

Why This Design?

- The CLS token aggregates global information (good for classification).
- Patch tokens retain spatial information (good for saliency).
- Both heads share the same backbone, so features learned for one task can help the other.
- The softmax on the saliency head ensures valid probability distributions by construction, which pairs naturally with KL divergence loss.

Datasets

- **SALICON**: fixation density maps for 10,000 train / 5,000 val COCO images.
- Collected via mouse-contingent gaze tracking (approximates eye tracking at scale).
- **COCO 2014**: bounding-box annotations for 80 object categories.
- We select the top 20 most frequent categories and create binary label vectors.



Example: image with ground-truth saliency overlay and XAI attributions.

Preprocessing

- Images resized to 224×224 , normalized with ImageNet statistics.
- No spatial augmentations (would need identical transforms on saliency maps).
- Saliency maps processed at two resolutions:
 - 224×224 (full resolution, for visualization)
 - 14×14 (patch grid, for training and evaluation)
- Grid-resolution maps normalized to sum to 1 (probability distribution over patches).

Training Setup

- **Optimizer:** AdamW, $\text{lr} = 10^{-4}$, weight decay = 10^{-4}
- **Schedule:** cosine annealing over 15 epochs
- **Batch size:** 32
- **Loss:** $\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda \cdot \mathcal{L}_{\text{sal}}$ with $\lambda = 1.0$
 - Classification: binary cross-entropy with logits
 - Saliency: KL divergence (or MSE for ablation)
- Best checkpoint saved by validation loss.
- Each training run takes 5–10 minutes on a single GPU.

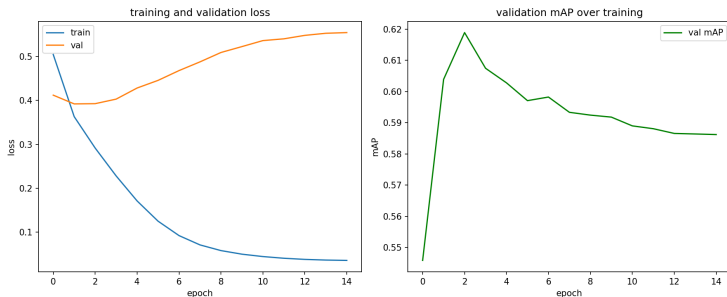
What We Compare

- **Frozen-feature baselines** (no fine-tuning):
 - Logistic regression and MLP on CLS token features (384-dim)
 - Mean saliency map, ridge regression, MLP regressor
- **Fine-tuned ablations:**
 - **multitask:** both heads active
 - **cls-only:** saliency loss zeroed out
 - **sal-only:** classification loss zeroed out
- **Loss ablation:** MSE vs KL divergence for saliency

Evaluation Metrics

- **Classification:** mean average precision (mAP) over 20 categories.
- **Saliency:**
 - CC (Pearson correlation): linear agreement with ground truth
 - SIM (histogram intersection): distribution overlap
 - KL divergence: distributional distance (lower is better)
- All saliency metrics computed on the 14×14 patch grid.

Training Curves



- Best validation loss at epoch 2–3 (mild overfitting after).
- Checkpoint selection prevents overfitting from affecting evaluation.

Classification Results

Method	mAP
Logistic regression (frozen)	0.556
MLP classifier (frozen)	0.566
ViT cls-only	0.623
ViT multitask (KL)	0.604

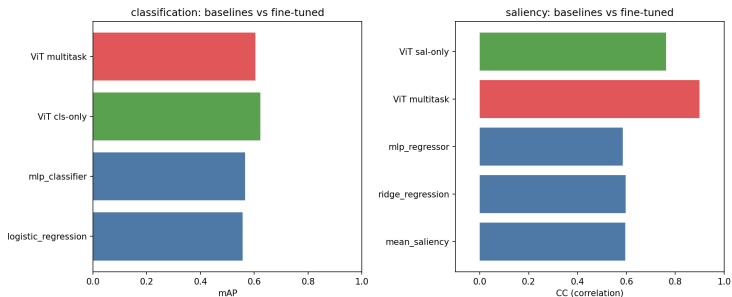
- Fine-tuning improves over frozen baselines by 5–6 mAP points.
- Multi-task pays a small cost (1.9 points) vs classification-only.
- This is the “price” of sharing the backbone with saliency.

The MSE vs KL Story

Saliency Loss	mAP	CC	SIM	KL
MSE	0.612	0.196	0.430	1.087
KL divergence	0.604	0.899	0.799	0.182

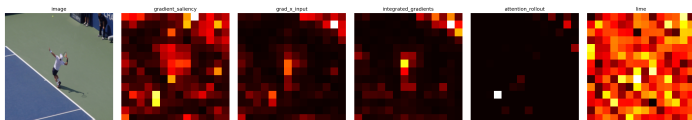
- Same architecture, same data, same optimizer, same λ .
- CC jumps from 0.196 to 0.899 just by switching the loss.
- MSE on softmax outputs does not encourage the distribution shape that CC/SIM measure.
- KL divergence directly compares probability distributions, aligning the loss with the evaluation metric.

Baseline Comparison (Visual)



- Left: classification mAP. Right: saliency CC.
- Fine-tuned models outperform frozen baselines on both tasks.

Saliency Prediction Example



A tennis player image. The model's saliency head learns to predict where humans look, concentrating on the player.

Key Takeaways

- ❶ **Multi-task learning works:** a single ViT can classify and predict saliency simultaneously with modest trade-offs.
- ❷ **Fine-tuning beats frozen features:** end-to-end adaptation extracts more task-relevant representations than sklearn baselines on frozen embeddings.
- ❸ **Loss design matters:** switching from MSE to KL divergence raises saliency CC from 0.196 to 0.899. Matching the loss to the output parameterization and evaluation metric is critical.
- ❹ **Multi-task trade-off is small:** only 1.9 mAP points lost for joint saliency prediction.

Limitations and Future Work

- Single training seed (42); multi-seed runs would strengthen conclusions.
- KL and MSE have different gradient magnitudes, so the effective task weighting also shifts; a λ sweep would help isolate the loss-alignment effect.
- Saliency evaluated at 14×14 resolution (not comparable to standard SALICON leaderboard scores).
- No spatial augmentation; saliency-aware augmentation could help.
- Future: λ ablation, higher-resolution saliency heads, cross-dataset evaluation.

Questions?

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Code available upon request.